

CoursePilot – Feature Brainstorming & MVP Scoping

Purpose of this document:

This document is meant for us to kick off team brainstorming around some potential product features for CoursePilot. Obviously at this point the goal here is idea generation (and maybe final decisions). We will use this document to converge on a realistic MVP scope that we can implement, demo, and defend.

Product Goal:

CoursePilot is an AI-powered student dashboard designed to reduce cognitive overload by converting unstructured academic materials (syllabi, slides, notes) into actionable schedules, reminders, and study support.

Our primary goal for the MVP is to:

Help a student upload a syllabus and immediately understand what matters, when it matters, and how to prioritize their work

Key Assumptions:

Before brainstorming features it is important to understand that we are operating under these assumptions:

1. Students are willing to upload syllabi manually
2. The Syllabi vary widely in format and in structure
3. AI should help with organizing and understanding, not academic dishonesty
4. Features we decide upon should be demo-able with 1-2 real courses

Core MVP Features (Possible Candidates):

A. Syllabus Upload & Parsing (Need to have)

- Upload syllabus PDF
- Our system extracts and stores syllabus text
- Page/section references are preserved for citations

Questions:

- Do we require syllabus upload before anything else?
- If a student already uploaded the same syllabus into our system do we offer them an opportunity to use the info already in our system?

B. Automatic Schedule Generation (Need to have)

- Extract Key dates for:
 - Exams
 - Quizzes
 - Demos
 - Assignments
- Display as:
 - Calendar-style
- Each item links back to the syllabus page it came from

Questions:

- Before the calendar is officially created do we show the user what was generated and they can edit themselves?

C. Notify Users About Upcoming Events: (Need to have)

- Notifications occur via:
 - In-app notifications
 - Email notifications
 - Browser push notifications
 - SMS text messages
- Notification timing options:
 - 1 week before
 - 3 days before
 - 24 hours before
 - Day of the event
- Display/Behavior:
 - Notifications reference the specific event (e.g., “Demo #1 due Feb 23”)
 - Each notification links back to the corresponding schedule item

Questions:

- Should notifications preferences apply to all courses or should we allow users to customize them per course
- Should notifications be enabled by default or opt-in?

D. Grading Breakdown Extraction (Nice to have)

- Automatically extract grading components and weights for:
 - Exams
 - Assignments
 - Team project / demos
 - Presentations

- Participation
- Display/Behavior:
 - Simple percentage breakdown (per course) with list or bar chart
 - Clear labels for each grading component
 - Each value links back to the syllabus page it was extracted from

Questions:

- Should grading components be editable by the user or locked in once they are extracted?
- How do we handle syllabi that have vague weights?

E. Assignment Checklist Generator (Nice to have)

- For each extracted event, generate a checklist of sub-tasks:
 - Example: “Research Paper” -> Outline, Draft, Peer Review, Final Edit, Submit
 - Allow users to mark sub-tasks as complete, add custom sub-tasks, and display progress percentage per assignment

Questions:

- Should AI auto-generate sub-tasks or provide templates the user customizes?
- Should we show aggregate progress across all courses?

F. Priority Dashboard / “What’s Next” View (Nice to have)

- Single page showing the next 3-5 most urgent items across all courses
- Could potentially color code by urgency (red = due soon, yellow = upcoming, green = far out)
- Quick-action buttons like mark as done, ignore, view details, etc.

Questions:

- How do we choose which items are the most “urgent”?
- Should our algorithms consider only due date proximity or include other things like grade weight and user-flagged importance?

G. Q&A Chatbot (Nice to have)

- Chat interface shown at the bottom of the screen (sort of like a “how can I help you today?”)
- Since the chatbot will have access to all the data provided by the user, users can ask things like “When is my first exam for CSCI-577A?” or “What percentage is the team project worth in CSCI-570?”
- Offers ability to answer user questions very quickly

Questions:

- Which LLM API would we want to utilize for this?
- Should we make sure chat history persists across the sessions?
- Do we limit the scope to just syllabi related questions or allow general study questions as well?
- How do we handle questions the Chatbot can't answer?

H. Grade Calculator (Nice to have)

- Based on extracted grading breakdown for a course, allow users to input the current or expected score per category
- Calculate current grade and projected final grade

Questions:

- Can we somehow connect actual grades to our product via Canvas/Brightspace?

I. Flashcard Generation from Notes & Slides (Nice to have)

- Auto-generate flashcards from uploaded slides/notes
- AI can potentially identify term and definition combinations, as well as general concept/explanation pairs
- User can review, edit, delete, or add cards before exporting
- Exported formats
 - CSV (manual import to Quizlet)
 - Anki package (.apkg) for spaced repetition
 - Printable PDF for physical flashcards

Questions:

- Should we limit auto-generated cards per upload? (e.g., max 25 or 50)?
- Should we support images on flashcards?
- Should card history persist in our system?

Notes: